	GLOBAL ECONOMICS **Grade: 10 - Learning Evidence 3** **DECISION-MAKING UNDER UNCERTAINTY** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Compare decision criteria and their limitations, apply staged filtering, and plan a justified investigation of a real-life economic or financial decision under uncertainty.	**Assessment criteria:** C8: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations. C9: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative. C10: Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

Answer individually on separate paper. Questions 1–6 are worth 2 points each; Question 7 has three components worth 2 points each. A **minimum of 4 points** is required to pass each criterion. Partial credit is available for correct comparisons, filtering stages, and justified reasoning even if later conclusions contain errors.

Question	C8	C9	C10
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	2	–
6	–	2	–
7(a), (b), (c)	–	–	$2+2+2$
Total	**6**	**6**	**6**

1. Repair Workshop Plans [C8-2]

A bicycle repair workshop chooses its capacity for the next quarter before demand is known. Profits are in thousand USD; forecast probabilities are shown below.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Large (A_1)	120	25	-20
Flexible (A_2)	85	65	10
Basic (A_3)	50	45	35
Partner (A_4)	70	75	20

The following table summarizes the decision-making statistics for each alternative.

	Max	Min	Regret	EMV
A_1	120	-20	55	35
A_2	85	10	35	58
A_3	50	35	70	44
A_4	75	20	50	63

(a) Report each criterion's recommended alternative(s), deciding statistic, and value. **[1]**

(b) A subsidy s (integer, $0 \le s \le 30$, thousand USD) increases the low-demand payoffs of both A_1 and A_2 by s. The rest remain unchanged. An alternative is eligible only if its profit is nonnegative in every state. Determine the ranges of s for which each alternative is eligible. Find the smallest s making A_2 the unique Maximin choice. At that s, report all four recommendations and deciding values among eligible alternatives. **[1]**

2. School Store Orders [C8-2]

A school store chooses one stationery order for the coming term. Net profits are in hundred USD.

Alternative	High	Normal	Low
$P(S_j)$	0.25	0.5	0.25
Order B_1	72	54	36
Order B_2	60	48	24
Order B_3	72	42	36

	Max	Min	Regret	EMV
B_1	72	36	0	54
B_2	60	24	12	45
B_3	72	36	12	48

(a) Report all four recommendations and deciding values, including every tie. **[1]**

(b) A supplier charges an extra flat fee f for B_1, deducted once from its profit in every state. Here f is an integer, $0 \leq f \leq 10$, in hundred USD. All other payoffs and probabilities stay fixed; all orders remain eligible. Find the smallest f making EMV/Bayes strictly prefer B_3 over B_1. Justify the threshold, identify the boundary tie, and interpret the fee in USD. At that f, report all four updated recommendations and deciding values, including every tie. **[1]**

3. Community Market Stall [C8-2]

An organizer chooses a stall for one weekend. Profits are in hundred USD. The forecast assigns high attendance probability 0.40 and low attendance probability 0.60.

Alternative	High	Low
$P(S_j)$	0.4	0.6
Large (C_1)	80	10
Compact (C_2)	48	32

	Max	Min	Regret	EMV
C_1	80	10	22	38
C_2	48	32	32	38.4

(a) For part (a) only, the forecast is withdrawn as unreliable. Compare the information needed by all four criteria: which summaries still support a choice, which no longer do, and why would equal probabilities need justification? **[1]**

(b) For this part, restore the supplied probabilities as reliable. A sponsor guarantees C_2 a high-attendance profit of g, replacing 48, where g is an integer, $48 \leq g \leq 80$, in hundred USD. All other payoffs stay fixed; both stalls remain eligible. Find the smallest g making C_2 the unique Minimax Regret choice. Justify the threshold and boundary tie; interpret the increase from 48 in USD. At that g, report all four updated recommendations and deciding values, including every tie. **[1]**

Staged filtering: Questions 4–6

For **each** problem, construct a compact filtering tree with branches for retained and eliminated alternatives. Label every stage with its rule and relevant numerical values. Identify the final choice, including any tie, and write one sentence interpreting the result in context. **[1]**

Also construct a supporting table with the following columns, recording every stage and its relevant values. **[1]**

# Step	Description	Left Alternatives	Eliminated Alternatives

- Step 0 includes all six alternatives.
- Each stage compares only survivors from the previous stage. Eliminated alternatives cannot reenter.
- Supplied summary statistics remain fixed throughout filtering. Do not recalculate regrets or other statistics after eliminations.
- When keeping the best k alternatives, retain all tied at the cutoff, even if more than k survive.
- Report any final tie explicitly; do not invent a tie-breaker.

Use highest EMV for EMV/Bayes, highest Max for Maximax, highest Min for Maximin, and lowest Regret for Minimax Regret. Regrets were calculated against all six original alternatives.

4. Book Fair Stock [C9-2]

A bookseller chooses a stock plan D_1–D_6 for a fair. Supplied summaries describe net profits in hundred USD under high, normal, and low sales, with probabilities 0.50, 0.30, and 0.20.

	EMV	Max	Min	Regret
D_1	65	100	0	40
D_2	65	90	10	30
D_3	65	80	20	20
D_4	59	70	30	30
D_5	53	60	40	40
D_6	43	50	30	50

Apply these stages in order:

1. EMV/Bayes: keep the 2 highest EMVs, retaining cutoff ties.
2. Maximin: select the highest Min among survivors.

Complete the tree, table, and interpretation described above.

5. Music Event Packages [C9-2]

An event organizer chooses one equipment package E_1–E_6 for a weekend music event. Summaries show net profits in hundred USD under high, normal, and low ticket sales, with probabilities 0.40, 0.40, and 0.20.

	EMV	Max	Min	Regret
E_1	58	110	-10	55
E_2	63	95	15	30
E_3	69	95	25	20
E_4	67	80	35	30
E_5	61	70	45	40
E_6	52	60	40	50

Apply these stages in order:

1. Maximax: keep the 3 highest Max values.
2. Maximin: keep the 2 highest Min values among survivors.
3. Minimax Regret: select the lowest Regret among survivors.

Construct the filtering tree and supporting table; include rules, values, eliminations, final choice, and a one-sentence interpretation. Use the shared rules for Questions 4–6.

6. Refill Shop Expansion [C9-2]

A refill shop chooses an expansion plan F_1–F_6 for the next year. Summaries show net profits in thousand USD under strong, steady, and weak demand, with probabilities 0.30, 0.50, and 0.20.

	EMV	Max	Min	Regret
F_1	61	120	0	45
F_2	64.5	105	15	30
F_3	68.5	95	25	25
F_4	70	85	35	35
F_5	64	75	45	45
F_6	55	65	40	55

Apply these stages in order:

1. Minimax Regret: keep the 5 lowest Regret values.
2. EMV/Bayes: keep the 4 highest EMVs among survivors.
3. Maximax: keep the 2 highest Max values among survivors.
4. Maximin: select the highest Min among survivors.

Construct the filtering tree and supporting table; include rules, values, eliminations, final choice, and a one-sentence interpretation. Use the shared rules for Questions 4–6.

7. Future Research Proposal [C10-6]

Write an **individual proposal** for the real-life economic or financial decision under uncertainty you intend to investigate for the C10 practice assignment. Organize your response into the three components below, using short, connected paragraphs.

This exam asks for your **research plan only**. You do not need completed research, collected data, a numerical payoff table, calculations, a final recommended alternative, completed references, or a PDF article. You may name intended sources; do not fabricate data or citations.

(a) Proposed decision and context. Describe a specific, realistic situation, the decision-maker, objective, and time period. Identify at least two feasible alternatives and at least two mutually exclusive states of nature covering the relevant possibilities. Explain the relevant risk attitude or constraints, such as budget or acceptable loss. **[2]**

(b) Planned evidence and financial model. Explain what financial information you will collect and where you expect to obtain it. Describe how revenues, costs, net gains, or savings will be used to construct a comparable payoff table in which larger values are better. State the intended currency, units, and time period. Explain how you will distinguish sourced facts from assumptions and estimates. **[2]**

(c) Proposed criterion and justification. Make a provisional choice among Maximax, Maximin, Minimax Regret, and EMV/Bayes. Explain why it fits the anticipated information, objective, risk attitude, and constraints. Briefly explain why each of the other three may be less suitable. **[2]**

If you propose EMV/Bayes, explain how you will obtain defensible probabilities for all states. Otherwise, explain why a criterion without probabilities fits. Identify one expected limitation or assumption that could affect the eventual recommendation.

Your criterion choice is **provisional** and may change after gathering evidence. The plan will be assessed for specificity, coherence, feasibility, and justification.